Vansh Sinha

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Education

Arizona State University **Tempe, AZ**

B.S. in Computer Science | Minor in Data Science | GPA: 3.8/4.0 *Aug. 2023 - May 2027*

- Relevant Coursework: Distributed Systems, Data Structures & Algorithms, Operating Systems, Software Engineering, Theory of Computing, Database Systems

Work Experience

American Express **June 2026 - Aug. 2026**

Software Engineer Intern | Global Loyalty Benefits *Phoenix, AZ*

- Designed and shipped a four-state deployment lifecycle to production for 100+ engineers, replacing destructive release deletes with auditable remove, revert, rollback, and active transitions.
- Built one-action release reactivation, replacing a multi-day redeploy and approval cycle for non-active releases.
- Delivered an event automation framework and rules UI end to end, generating 250+ weekly Slack alerts for release milestones, unapproved RFCs, and missed check-ins.
- Engineered scheduler reliability with retries, idempotency, backoff, and job-state logging; added unit and integration tests.
- Presented the deployment lifecycle and automation framework to senior leadership; received a return offer.
- Used AI coding tools daily with persistent repository context and Markdown sprint trackers; validated plans and changes through mentor 1:1s and GitHub code reviews.

Banner University Medical Center Phoenix **Aug. 2026 - Present**

Software Developer, ASU Senior Capstone Team | Cardiac Electrophysiology *Phoenix, AZ*

- Working on a four-engineer team building an EKG-image application for cardiac arrest risk scoring for Banner Health's Cardiac Electrophysiology group.
- Building an ECG digitization pipeline to convert paper and on-screen EKG images into digital waveforms and extract T-wave attributes for risk scoring.
- Developing EKG image de-identification for retrospective patient-data analysis and a fallback capture screen with image cropping for cases outside the primary capture SDK.

Wimoku **June 2025 - Aug. 2025**

Software Engineer Intern | Data Infrastructure *Remote, USA*

- Built a real-time financial data pipeline with Pub/Sub and Apache Beam, streaming market data into BigQuery for analytics.
- Replaced managed Dataflow with Pub/Sub and Cloud Functions, reducing pipeline cost 60% while maintaining throughput.
- Shipped a Dockerized Kubernetes CronJob service executing 200+ audited BigQuery queries daily, replacing manual runs.

Projects

SaveMyGrade | Next.js, React, TypeScript, FastAPI, Gemini, ChromaDB, SQLite **GitHub**

- Built a local-first study platform that unifies PDFs, images, text, videos, and YouTube content into a searchable knowledge base with vector-embedding semantic search and grounded RAG chat.
- Created an interactive, force-directed knowledge graph from Gemini-extracted concepts, linking visual exploration directly to source material.
- Engineered asynchronous ingestion with persistent job states, concurrent uploads, cancellation, and crash recovery through startup orphan-job detection.
- Built source-cited flashcards with staleness detection and safeguards that preserve existing decks when regeneration fails.

TrackDeck | Swift, SwiftUI, AppKit, macOS Accessibility APIs **2026**

- Built a native macOS productivity app with a custom gesture engine that maps trackpad taps, pressure, and directional gestures to commands for the active application.
- Implemented workflow orchestration that turns one gesture into multi-step developer actions, including opening a terminal, initializing Git, creating files, and launching tools.

Leadership & Activities

Hacker Devils, ASU **Aug. 2023 - Present**

- Placed in the top 3 in two coding competitions.

AI Society, ASU **Aug. 2024 - Present**

- Built AI applications with peers in the AI Maker Space; studied ML fundamentals, NLP, LangChain, and RAG.

Technical Skills

Languages: Python | Go | Java | C/C++ | Swift | SQL | TypeScript | HTML/CSS
Frameworks: React | Next.js | FastAPI | Node.js | Tailwind CSS | SwiftUI | Apache Beam
Data & Databases: PostgreSQL | SQLite | BigQuery | Redis | ChromaDB | Prisma | pandas | NumPy
Cloud & Infrastructure: Docker | Kubernetes | GCP | Pub/Sub | Dataflow | Cloud Functions | Prometheus | Git
Concepts: RAG | Semantic Search | Vector Embeddings | Event-Driven Automation | Streaming Pipelines | Distributed Systems